FastSpeech 2-Based English and Telugu Text-to-Speech Synthesis

Technical Report

This report documents the implementation, dataset construction, training behaviour, inference outputs, variance-adaptor predictions, limitations, and reproducibility of two independent FastSpeech 2-based text-to-speech systems.

Executive summary

The project trains separate grapheme-based FastSpeech 2 models for English and Telugu. English uses ai4bharat/Svarah and Telugu uses SPRINGLab/IndicTTS_Telugu. Each language contains approximately five hours of training audio and one hour of validation audio, with zero manifest-ID overlap between the two splits. Both models completed ten epochs, and all five requested test sentences per language have generated WAV files and duration, pitch, and energy plots.

1. Implementation approach and architecture

The implementation follows the FastSpeech 2 design: a non-autoregressive text-to-mel model with explicit duration, pitch, and energy conditioning. The reference architecture was adapted for a self-contained two-language project and for the available data representation.

Architecture components

• Grapheme/character embedding and transformer encoder.

• Variance adaptor with duration, pitch (F0), and energy predictors.

• Length regulator that expands token representations according to duration values.

• Transformer mel decoder and 80-bin mel projection.

• Masked multi-task objective combining mel reconstruction, log-duration, pitch, and energy losses.

• Griffin-Lim vocoder for WAV synthesis without requiring an external neural vocoder.

Modifications for this project

• Grapheme tokenization is used instead of a phoneme front end. Telugu Unicode characters are retained without phonemization.

• A deterministic uniform frame-duration alignment is used as a baseline because forced-alignment annotations are not supplied.

• English and Telugu have independent configurations, vocabularies, manifests, feature directories, checkpoints, and result directories.

• The inference entry points preserve the supplied test text and produce WAV audio, mel arrays, variance CSV files, and variance-adaptor plots.

2. Dataset selection, subsampling, and preprocessing

Dataset and split statistics

Language	Dataset / source split	Training set	Validation set	Overlap
English	ai4bharat/Svarah / auto-selected	3,392 records; 18,021.95 s (5.006 h)	721 records; 3,609.39 s (1.003 h)	0 IDs

Language	Dataset / source split	Training set	Validation set	Overlap
Telugu	SPRINGLab/IndicTTS_Telugu / train	891 records; 18,013.36 s (5.004 h)	162 records; 3,614.00 s (1.004 h)	0 IDs

Selection and split policy

• Load only the configured language dataset and source split.

• Require non-empty text and decodable audio; skip audio shorter than 0.3 seconds.

• Measure actual source-audio duration, sort by stable record ID, and shuffle with deterministic seed 1234.

• Accumulate complete utterances to 18,000 seconds for training and 3,600 seconds for validation. Utterances are not truncated, so totals can slightly exceed targets.

• Construct validation only from records not selected for training. Independent verification found zero shared IDs for English and zero shared IDs for Telugu.

Preprocessing

• Decode audio, convert to mono, and resample to 22,050 Hz.

• Extract 80-bin mel spectrograms with n_fft=1024, hop_length=256, f_min=0, and f_max=8000.

• Extract pitch/F0 and energy features.

• Normalize and tokenize text at grapheme/character level; Telugu Unicode graphemes are preserved.

• Assign uniform token durations including BOS/EOS, save compressed feature arrays, and write separate JSONL manifests.

• Build each vocabulary from that language's training transcripts only.

3. Training configuration and loss-curve analysis

Both models use hidden size 192, four transformer layers, two attention heads, batch size 4, AdamW with learning rate 0.0001 and weight decay 0.01, and ten epochs. The saved checkpoints confirm epoch 10 for both languages.

Language	Epochs	Final train loss	Final validation loss	Best validation epoch
English	10/10	5283.1335	5380.0744	6; 5374.2705
Telugu	10/10	7269.1571	7778.9179	8; 7772.6610

English loss behaviour

The English training loss decreases from 5308.46 at epoch 3 to 5283.13 at epoch 10. Validation loss reaches its minimum at epoch 6 (5374.27), then oscillates mildly between approximately 5375 and 5397 while the final value is 5380.07. This indicates broad convergence with small epoch-to-epoch noise rather than a strong divergence. The gap between final training and validation losses is modest relative to their scale; there is no clear evidence of severe overfitting in the recorded curve, although the short ten-epoch run limits the strength of that conclusion.

Telugu loss behaviour

The Telugu training loss decreases gradually from 7308.10 at epoch 5 to 7269.16 at epoch 10. Validation loss is lowest at epoch 8 (7772.66), rises at epoch 9 (7786.91), and finishes at 7778.92. The curve therefore shows convergence with mild validation instability. The persistent validation-training gap is larger than for English, which may reflect differences in dataset size, language character distribution, feature statistics, or the uniform-duration baseline. It is not sufficient by itself to diagnose overfitting, but it motivates longer training and improved alignment for Telugu.

The corresponding plots are stored at `Results/English/` and `Results/Telugu/` as training_loss.png and validation_loss.png.

4. Generated speech observations

All five requested sentences were synthesized for each language using the corresponding best checkpoint. Audio duration is an objective output measurement; pronunciation and naturalness require human listening evaluation, which was not performed automatically in this verification. The observations below therefore distinguish verified artifacts from expected test-case challenges and avoid claiming intelligibility that cannot be established from file existence alone.

English test sentences

• **E1**: The long technical word stresses long-sequence handling and unusual grapheme context; it produced an 8.95 s WAV.

• **E2a**: This sentence tests the noun/verb ambiguity of 'conduct' in a natural context and produced a 6.03 s WAV.

• **E2b**: This sentence uses the same ambiguous word in a different syntactic context and produced a 6.27 s WAV.

• **E3**: The supplied text intentionally contains the spaced form 'rhyth m'; the inference pipeline preserves the supplied text and produced a 4.24 s WAV. This is a known text-quality challenge rather than a model correction.

• **E4**: The sentence combines time, date, mixed-case text, and the nonstandard token 'andaroo'; it produced a 7.41 s WAV and stresses normalization and grapheme coverage.

Telugu test sentences

• **T1**: The long technical sentence stresses Telugu grapheme coverage and long-sequence duration; it produced a 6.54 s WAV.

• **T2a**: This short sentence tests the first context for the word 'pattu' and produced a 3.38 s WAV.

• **T2b**: This short sentence tests the contrasting clothing-related context for the same written form and produced a 3.94 s WAV.

• **T3**: This short, simple sentence provides a relatively controlled pronunciation case and produced a 3.29 s WAV.

• **T4**: The sentence combines a date, time, punctuation, and a long clause; it produced an 8.67 s WAV and stresses number and punctuation handling.

For a formal quality claim, each WAV should be listened to by native or expert evaluators and scored for intelligibility, pronunciation, prosody, and naturalness. The current report records reproducible generation and objective output properties rather than inventing subjective scores.

5. Variance-adaptor analysis

The inference pipeline writes one duration, pitch/F0, and energy plot per sentence and corresponding CSV files. The following statistics summarize the plotted predictions. Because duration alignment is initialized uniformly, the observed duration ranges are narrow and regular; this is expected from the baseline and should not be interpreted as learned phonetic alignment quality.

English predictions

ID	Audio	Tokens	Duration frames	Pitch mean / range	Energy mean / range
E1	8.95 s	110	7-8	175.7 / 147.9-176.5	-3.96 / -5.19--3.92
E2a	6.03 s	74	7-8	175.5 / 147.9-176.5	-3.97 / -5.19--3.92
E2b	6.27 s	77	7-8	175.5 / 147.9-176.5	-3.96 / -5.19--3.92
E3	4.24 s	52	7-8	175.2 / 147.9-176.5	-3.98 / -5.19--3.92
E4	7.41 s	91	7-8	175.6 / 147.9-176.5	-3.96 / -5.19--3.92

Across E1-E4, English token durations are 7-8 frames, mean pitch is approximately 175 Hz, and mean energy is approximately -3.96 in the stored feature scale. The similar ranges across sentences show consistent baseline conditioning. The pitch range includes a lower value around 148 Hz and an upper region around 176 Hz, indicating some predicted contour variation, but the compact range suggests limited expressive prosody. Energy is similarly stable, so emphasis differences may be understated.

Telugu predictions

ID	Audio	Tokens	Duration frames	Pitch mean / range	Energy mean / range
T1	6.54 s	70	8-10	188.6 / 78.9-197.0	-4.23 / -7.41--4.04
T2a	3.38 s	36	8-10	183.1 / 71.6-196.4	-4.34 / -7.32--4.04
T2b	3.94 s	42	8-10	185.1 / 73.6-197.2	-4.32 / -7.34--4.04
T3	3.29 s	35	8-10	182.7 / 71.4-196.3	-4.34 / -7.31--4.02
T4	8.67 s	93	8-10	189.7 / 81.1-197.0	-4.20 / -7.44--4.04

Across T1-T4, Telugu token durations are 8-10 frames, mean pitch ranges from approximately 182.7 to 189.7 Hz, and mean energy ranges from approximately -4.34 to -4.21. The wider pitch ranges, including values near 72-81 Hz and near 196-197 Hz, show more contour spread than English in these predictions. T1 and T4 have the highest mean pitch, consistent with longer sentences having more opportunities for contour variation, although this is not a linguistic prosody evaluation. Energy remains compact across all Telugu sentences, suggesting that the baseline model may produce limited phrase-level emphasis.

Per-sentence interpretation

• E1 and T1: long technical sentences have the largest frame counts and longest audio, so they are the most demanding tests of sequence length and coverage.

• E2a/E2b and T2a/T2b: paired sentences test contextual ambiguity. The plotted predictions and audio generation confirm that both contexts are processed independently, but objective plots cannot establish whether the semantic pronunciation distinction is correct.

• E3 and T3: short sentences make duration regularity easy to inspect. E3 additionally contains the supplied spacing anomaly, which is intentionally preserved.

• E4 and T4: dates, times, punctuation, and long clauses expose limitations in grapheme-only normalization and number pronunciation. Dedicated text normalization or a phoneme front end would likely improve these cases.

6. English-Telugu comparison, limitations, and improvements

• English has more training records (3,392 versus 891) for nearly the same duration target, while Telugu has fewer, longer utterances. This can affect optimization noise and generalization.

• English validation loss is lower in absolute value, but losses are not directly comparable across languages because vocabularies, feature distributions, and sequence statistics differ.

• Telugu outputs show a somewhat wider predicted pitch range in the stored statistics, while both languages show narrow duration and energy ranges characteristic of the uniform-duration baseline.

• The Griffin-Lim vocoder is self-contained and reproducible but can sound robotic or contain phase artifacts compared with a neural vocoder.

• Uniform token durations are not forced alignments. Montreal Forced Aligner or a learned alignment method would improve duration supervision and prosody.

• A phoneme or language-aware front end, explicit number/date normalization, native-speaker evaluation, and more epochs or model capacity are possible improvements.

• The current data and plots support reproducibility and objective inspection, but they do not provide MOS, intelligibility, word-error rate, or human pronunciation scores.

7. Reproducibility

The project uses Python with NumPy, PyYAML, PyTorch, librosa, soundfile, matplotlib, tqdm, Hugging Face datasets, pytest, and ReportLab for the generated report. The authoritative language configurations are `English/config.yaml` and `Telugu/config.yaml`.

• Install dependencies: `python -m pip install -r requirements.txt`.

• Preprocess English: `python English\preprocess.py --config English\config.yaml`.

• Preprocess Telugu: `python Telugu\preprocess.py --config Telugu\config.yaml`.

• Train English: `python English\train.py --config English\config.yaml`.

• Train Telugu: `python Telugu\train.py --config Telugu\config.yaml`.

• Run English inference: `python English\inference.py --config English\config.yaml --checkpoint English\checkpoints\best.pt`.

• Run Telugu inference: `python Telugu\inference.py --config Telugu\config.yaml --checkpoint Telugu\checkpoints\best.pt`.

• Run tests: `python -m pytest -q`.

The checked output includes 10/10 completed epochs for each language, 10 generated WAV files, 30 variance-adaptor plots, separate loss plots, and a zero-overlap train/validation verification for both datasets.

Conclusion

The implementation meets the requested two-language FastSpeech 2 workflow and submission artifacts. It provides reproducible duration-based dataset construction, independent English and Telugu training, complete inference outputs, and documented objective analysis. The principal limitations are the uniform duration baseline, Griffin-Lim synthesis, limited ten-epoch training, grapheme-only text handling, and the absence of human listening scores. These limitations are explicitly identified so future evaluation can distinguish verified pipeline completion from subjective speech-quality claims.

Objective Audio-Quality Addendum

This addendum supplements the Technical Report with reproducible checks of the ten generated WAV files. The checks verify file-level and signal-level properties; they do not establish pronunciation correctness, intelligibility, naturalness, or linguistic prosody.

ID	Duration (s)	Rate	Channels	Peak	RMS	Near-silence	Clipped samples
E1	8.95	22050	mono	0.9500	0.1701	0.05%	0
E2a	6.03	22050	mono	0.9500	0.2159	0.03%	0
E2b	6.27	22050	mono	0.9500	0.2197	0.03%	0
E3	4.24	22050	mono	0.9500	0.2021	0.04%	0
E4	7.41	22050	mono	0.9500	0.2186	0.03%	0
T1	6.54	22050	mono	0.9500	0.2136	0.04%	0
T2a	3.38	22050	mono	0.9500	0.2389	0.05%	0
T2b	3.94	22050	mono	0.9500	0.2129	0.05%	0
T3	3.29	22050	mono	0.9500	0.1911	0.06%	0
T4	8.67	22050	mono	0.9500	0.1970	0.04%	0

Objective findings. All files decode successfully as mono 22,050 Hz WAV audio. Peak amplitudes are approximately 0.95, with zero samples at or above the clipping threshold used here (0.999). Near-silence is below 0.1% for every file, so these checks found no clipping or truncated-silence anomaly.

Pronunciation and speech-quality status. Human listening observations were not available for this revision. Consequently, no unsupported claim is made about pronunciation, intelligibility, naturalness, prosody, robotic quality, or audible artifacts. Evaluators should listen to E1-E4 and T1-T4 and record those judgments separately. The sentence-specific test challenges and generated durations documented in the main report remain valid objective observations.

Reproducibility. The metrics were computed with SoundFile and NumPy using the WAV files in Results/English/test_generated_audios/ and Results/Telugu/test_generated_audios/.